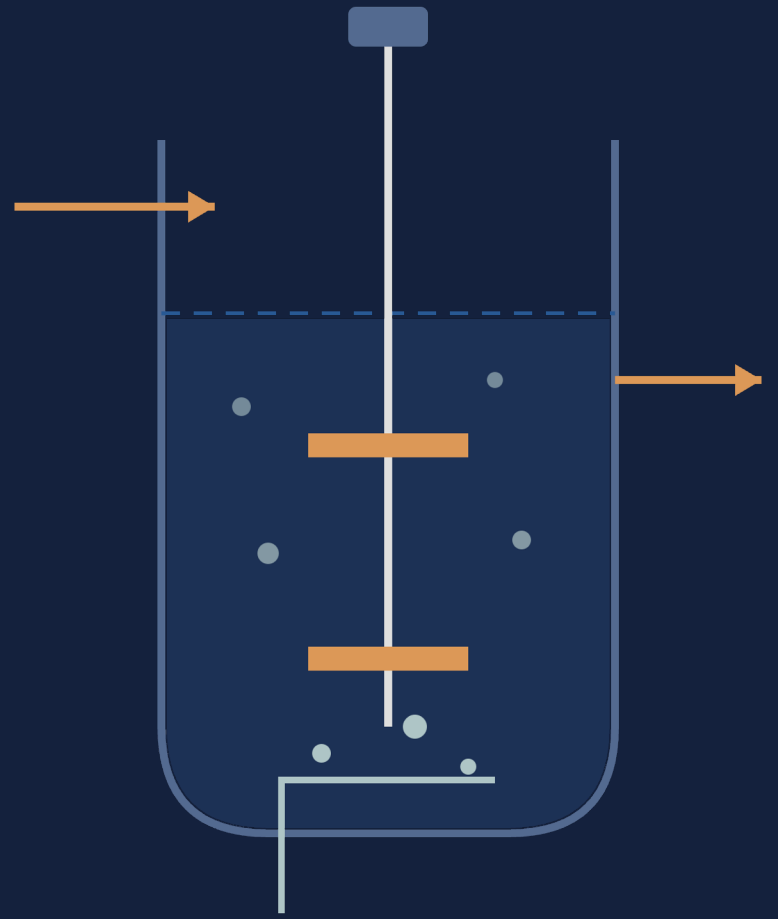
CL 420 Biochemical Engineering

Design of Fermentation Processes



Choosing the reactor and its operating conditions to get the best productivity, yield and final titer

Based on Villadsen, Nielsen & Lidén, *Bioreaction Engineering Principles*, 3rd ed., Springer (2011) Chapter 9

INTRODUCTION

What design optimises — the COGS

Productivity (rate)

kg product per m³ reactor per hour — how well the capital is used

Yield

g product per g substrate — the raw-material cost

Final titer

g product per litre — the downstream processing cost

Lecture roadmap

1. 9.1 Steady-state CSTR: productivity, coupled tanks, cell recycle, gas substrates
2. 9.2 Batch and fed-batch operation
3. 9.3 Transients, stability and mixed cultures in a CSTR
4. 9.4 Plug flow and loop reactors

Two worked numerical examples: CSTR optimum and exponential fed-batch

REACTOR TYPES · TABLE 9.1

Reactor types and operating modes

Mode	Strengths	Weaknesses	Typical use
CSTR (steady state)	Highest productivity; low capital cost; steady quality	Infection or strain mutation over long runs; inflexible	Cheap bulk products, waste water, insulin campaigns
Batch	Flexible between products; easy to sterilise; short runs	Idle time for cleaning, sterilising and inoculum growth; labour	Multi-product plants
Fed-batch	Avoids overflow metabolism and catabolite repression	Lower productivity than CSTR; large volume to harvest	Baker's yeast, proteins, antibiotics; **Biologics**
PFR / loop	Very high conversion; efficient use of gas substrates	Needs cells in the feed, so only after a CSTR	**Gas Fermentations**; SCP from CH_4 or CH_3OH

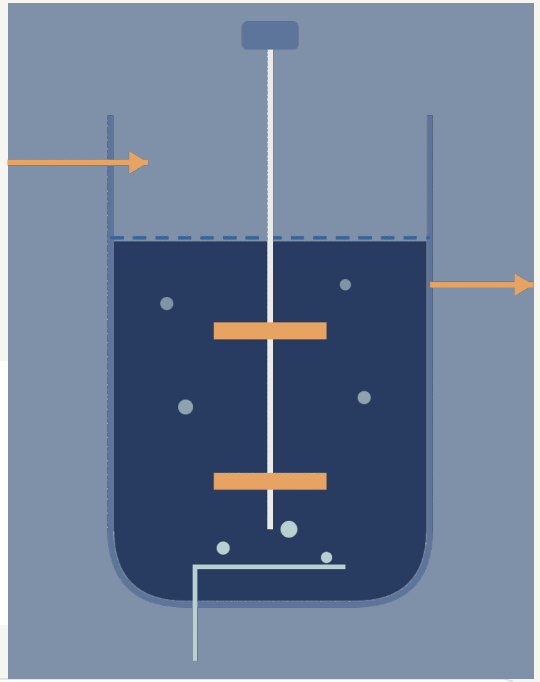
The stirred tank (STR) is by far the most common bioreactor. Mixing is assumed ideal throughout Chapter 9.

THE STARTING POINT · EQ. 9.1

One mass balance covers every tank

$$\frac{d(Vc)}{dt} = V\,(q_t + q) + v_f\,c_f - v_e\,c_e$$

q_t : gas-to-liquid transfer rate q : net volumetric production rate f, e : feed, effluent



CSTR

Steady state, $v_f = v_e = v$, $D = v/V$

$$q_t + q + D\,(c_f - c) = 0$$

Batch

No flows in or out, V constant

$$\frac{dc}{dt} = q(c)$$

Fed-batch

Feed only ($v_e = 0$), $V = V(t)$

$$\frac{dc}{dt} = q(c) + \frac{v}{V}\,(c_f - c)$$

Batch and fed-batch cultures stay near **metabolic steady state**, so Monod kinetics still apply.

In a chemostat, D sets the growth rate

Monod kinetics, sterile feed, no maintenance

$$D = \mu = \frac{\mu_{\max}\, s}{K_s + s}$$

$$s = \frac{K_s\, D}{\mu_{\max} - D}$$

$$x = Y_{x/s}\,(s_f - s)$$

$$p = Y_{p/s}\,(s_f - s)$$

Washout: $D_{\max} = \dfrac{\mu_{\max}\, s_f}{K_s + s_f}$

With maintenance

$$D\,(s_f - s) = \left(\frac{D}{Y^{\text{true}}_{x/s}} + m_s\right) x$$

- Biomass falls to zero both at very low D and at washout, with a maximum in between
- As D → 0, all substrate is burnt for maintenance (to CO_2)
- Example 9.1 (glycerol): biomass peaks at D = 0.866 h^{-1}

Maximising biomass productivity

Monod, no maintenance: $a = K_s/s_f, \quad S = s/s_f$

$$q_x = D\,x = \mu_{\max}\,Y_{x/s}\,s_f\,\frac{S\,(1-S)}{S+a}$$

$$S_{\text{opt}} = -a + \sqrt{a^2 + a}$$

$$D_{\text{opt}} = \mu_{\max}\left(1 - \frac{1}{\sqrt{1+1/a}}\right)$$

$$(q_x)_{\max} = \mu_{\max}\,Y_{x/s}\,s_f\left(\sqrt{1+a} - \sqrt{a}\right)^2$$

The optimum is never at...

- the lowest s — there μ, and so D, is near zero
- the highest biomass or product concentration
- the highest yield

When s equals the feed concentration there is no biomass; at s = 0 there is no growth. Productivity peaks in between, usually close to washout.

Kinetics decide where the optimum lies

Base case: $\mu_{max} = 0.5\ \mathrm{h^{-1}}$, $K_s = 0.4\ \mathrm{g\,L^{-1}}$, $s_f = 2\ \mathrm{g\,L^{-1}}$, $Y_{x/s} = 0.5\ \mathrm{g\,g^{-1}}$

Rate expression	Optimal residence time
Monod	3.37 h
Substrate inhibition	7.17 h
Product inhibition	4.69 h
Monod + maintenance	4.29 h

Any inhibition or maintenance lengthens the residence time, so the reactor gets bigger for the same output.

Substrate inhibition: a trap

$$\mu = \frac{\mu_{max}\, s}{K_s + s + s^2/K_i}$$

μ peaks at $s = \sqrt{K_i\, K_s}$

Steady states beyond the peak are unstable: a small rise in s leads to washout. Operate well to the left of it — never at the maximum μ.

WORKED EXAMPLE 1 · PROBLEM

Sizing up a 1 m³ chemostat

Given

A bacterium grows on glucose in a 1 m³ CSTR; sterile feed, Monod kinetics, no maintenance.

Parameter	Symbol	Value
Max. growth rate	$\mu_{\max}$	0.5 h^{-1}
Saturation constant	K_s	0.2 g L^{-1}
Glucose in feed	s_f	10 g L^{-1}
Biomass yield	$Y_{x/s}$	0.5 g g^{-1}

Find

(a) the substrate, biomass and productivity at D = 0.3 h^{-1}

(b) the washout dilution rate

(c) the dilution rate giving maximum biomass productivity, that productivity, and the feed flow rate

Try it before the next slide: which answer do you expect to be closest to washout?

Step by step

(a) D = 0.3 h⁻¹

$$s = \frac{0.2 \times 0.3}{0.5 - 0.3} = \mathbf{0.30}\ \mathrm{g\,L^{-1}}$$

$$x = 0.5\,(10 - 0.3) = \mathbf{4.85}\ \mathrm{g\,L^{-1}}$$

$$q_x = D\,x = 0.3 \times 4.85 = \mathbf{1.46}\ \mathrm{g\,L^{-1}\,h^{-1}}$$

(b) Washout

$$D_{\max} = \frac{0.5 \times 10}{10 + 0.2} = \mathbf{0.49}\ \mathrm{h^{-1}}$$

(c) Maximum productivity

$$a = K_s / s_f = 0.02$$

$$S_{\mathrm{opt}} = -0.02 + \sqrt{0.0004 + 0.02} = 0.123$$

$$s = 0.123 \times 10 = \mathbf{1.23}\ \mathrm{g\,L^{-1}}$$

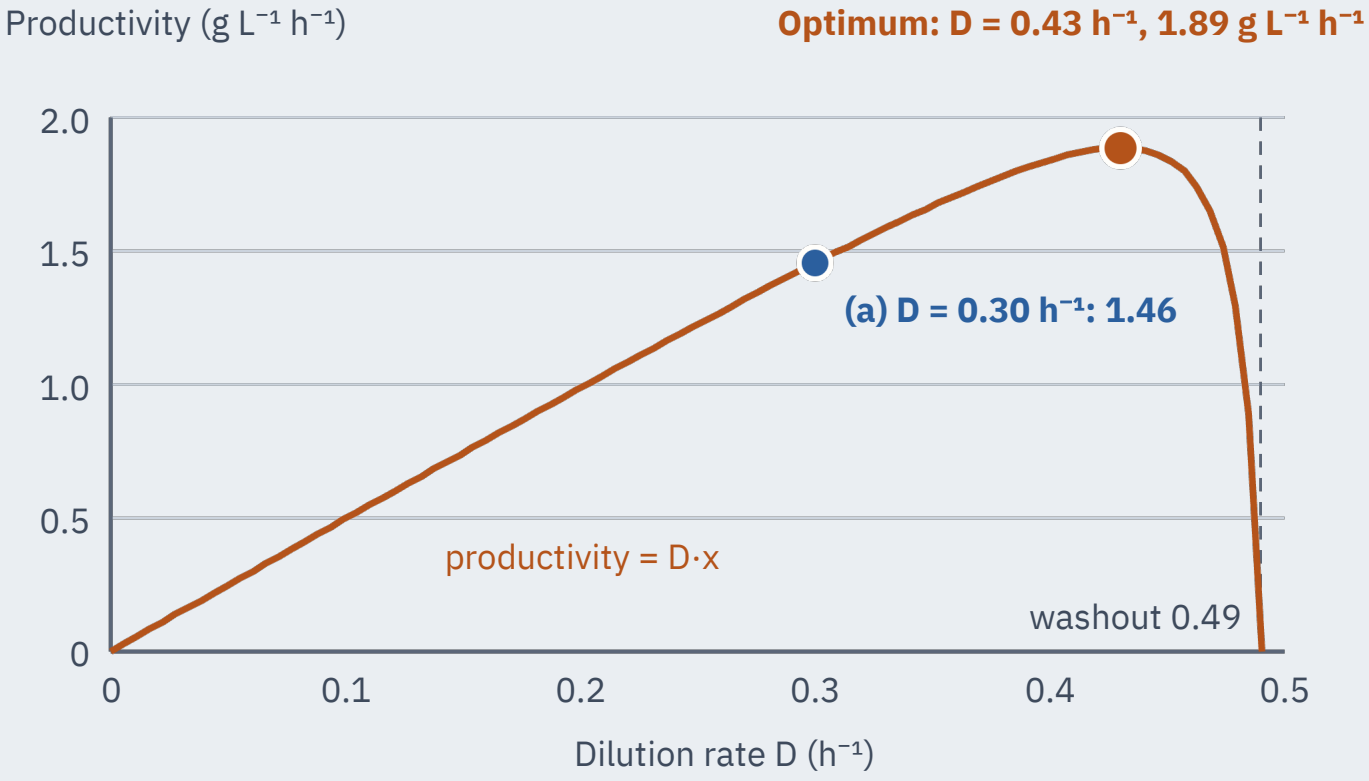
$$x = 0.5\,(10 - 1.23) = \mathbf{4.39}\ \mathrm{g\,L^{-1}}$$

$$D_{\mathrm{opt}} = 0.5\left(1 - 1/\sqrt{51}\right) = \mathbf{0.43}\ \mathrm{h^{-1}}$$

$$(q_x)_{\max} = 0.43 \times 4.39 = \mathbf{1.89}\ \mathrm{g\,L^{-1}\,h^{-1}}$$

$$v = D_{\mathrm{opt}}\,V = \mathbf{0.43}\ \mathrm{m^3\,h^{-1}}$$

Productivity peaks just before washout



+30% output

Moving from D = 0.30 to 0.43 h^{-1} lifts productivity from 1.46 to 1.89 g L^{-1} h^{-1}.

88% of washout D

The optimum sits close to washout, so feed disturbances are risky.

12% substrate lost

Effluent s rises from 0.30 to 1.23 g L^{-1}: productivity is bought with yield.

Why one tank fails in a storm

$$\mu = \frac{4}{3}\,\frac{s}{s+4}\ \mathrm{h^{-1}}\ \ (s \text{ in mg L}^{-1}), \qquad Y_{x/s} = 0.1, \qquad s_f = 60\ \mathrm{mg\,L^{-1}}$$

1 · Run the tank at maximum productivity

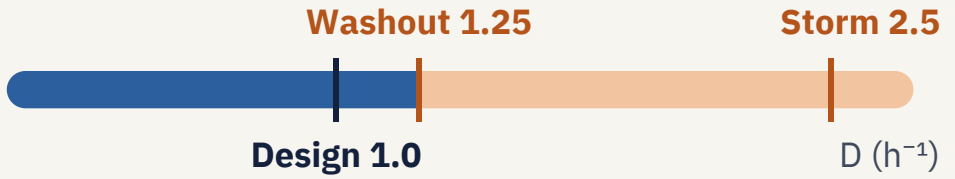
$$S_{\mathrm{opt}} = 0.2 \ \Rightarrow\ s = 12\ \mathrm{mg\,L^{-1}}, \qquad D_{\mathrm{opt}} = \frac{4}{3}\cdot\frac{12}{12+4} = 1\ \mathrm{h^{-1}}$$

2 · Size it: 1 m³ h⁻¹ of waste water needs a 1 m³ tank

$$q_s = D_{\mathrm{opt}}\,(s_f - s) = 1 \times (60 - 12) = 48\ \mathrm{mg\,L^{-1}\,h^{-1}}$$

3 · Find the washout limit

$$D_{\max} = \frac{\mu_{\max}\, s_f}{K_s + s_f} = \frac{4}{3}\cdot\frac{60}{60+4} = 1.25\ \mathrm{h^{-1}}$$



4 · Then the storm arrives

$$D = \frac{v}{V} = \frac{2.5\ \mathrm{m^3\,h^{-1}}}{1\ \mathrm{m^3}} = 2.5\ \mathrm{h^{-1}} > D_{\max}$$

Cells are washed out twice as fast as they can grow, and the sludge takes a long time to regrow. A more concentrated feed is not the danger — s stays at 12 mg L⁻¹ and only biomass rises. More flow is.

Three ways to add a second tank

(a) In parallel



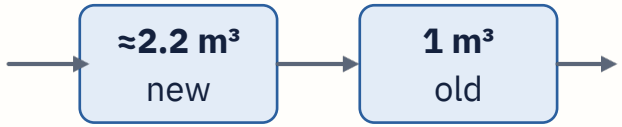
Add a 1.5 m³ tank and split the flow 1.5 : 1. Each tank runs at D = 1 h⁻¹, its optimum.

48 mg L⁻¹ h⁻¹

removed · effluent 12 mg L⁻¹

(b) In series

All 2.5 m³ h⁻¹ through both tanks

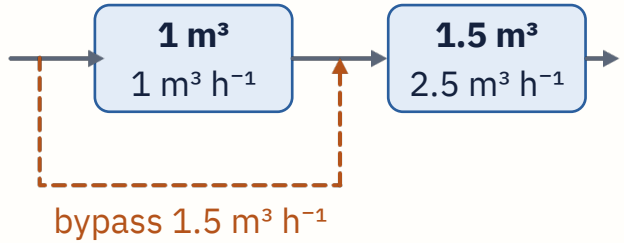


The new tank seeds the old one with cells, so the old tank survives D = 2.5 h⁻¹. Cleaner effluent, but more total volume.

43 mg L⁻¹ h⁻¹

removed · effluent ≈ 5 mg L⁻¹

(c) Series + split feed



The small tank runs at its optimum and seeds the large tank, which cleans the mixed stream.

48 mg L⁻¹ h⁻¹

removed · effluent 12 mg L⁻¹ · best in bigger storms

Batch design: few knobs, simple models

$$\frac{dx}{dt} = \mu\, x \qquad s = s_0 - \frac{x - x_0}{Y_{x/s}}$$

$$p = Y_{p/x}\,(x - x_0)$$

Valid for any μ while yields are constant. Integrated for Monod kinetics:

$$\mu_{\max} t = \left(1 + \frac{a}{1 + X_0}\right) \ln \frac{X}{X_0} - \frac{a}{1 + X_0} \ln(1 + X_0 - X)$$

$$X = \frac{x}{Y_{x/s}\, s_0}, \qquad a = \frac{K_s}{s_0}$$

Two asymptotes

Early: $x \approx x_0\, e^{\mu_{\max} t}$ Late: $x \to x_0 + Y_{x/s}\, s_0$

A low initial substrate lets a single batch give both kinetic parameters.

Substrate inhibition

Growth looks like a long lag phase. For toxic substrates, prefer a steady-state CSTR.

Product inhibition

Explains “stuck” wine fermentations that stop with sugar left over.

Why feed the substrate slowly?

Overflow metabolism
High sugar gives ethanol in yeast, acetate in E. coli

Catabolite repression
Product synthesis switched off at high sugar

Substrate inhibition
Batch starts poorly at high s_0

Finite run time
Less risk of mutation or infection than a CSTR

$$\frac{dc}{dt} = q(c) + \frac{v}{V}(c_f - c)$$

$$\frac{dV}{dt} = v(t)$$

Like a CSTR, but D = v/V changes with time

Two feed policies

Constant μ (s held constant): exponential feed, fastest growth without by-products.

Constant volumetric growth rate: once O_2 transfer or cooling limits the rate.

Start from three mass balances

Assumptions

- Feed v(t) carries sugar but no cells
- Nothing is withdrawn until harvest
- No maintenance: all sugar consumed becomes biomass, with yield Y
- Feed is controlled so sugar stays at $s = s_0$. Under Monod, μ depends only on s, so $\mu = \mu_0$ stays constant

Why balance xV and sV? They are total grams in the tank. Concentrations alone mislead because V keeps growing.

Volume

Nothing leaves, so V grows at the feed rate.

$$\frac{dV}{dt} = v$$

Biomass

No cells enter or leave. Total biomass only grows.

$$\frac{d(xV)}{dt} = \mu\, xV$$

Substrate

Sugar enters with the feed. Each gram of new cells uses 1/Y grams.

$$\frac{d(sV)}{dt} = v\, s_f - \frac{\mu}{Y_{x/s}}\, xV$$

PART A · SLIDE 15 · FED-BATCH AT CONSTANT M

Steps 1–2: biomass first, then the feed rate

Step 1 · Biomass balance

With μ = μ_0 fixed, total biomass grows first-order. Separate variables and integrate from x_0V_0 at t = 0:

$$xV = x_0 V_0 \, e^{\mu_0 t}$$

This is the slide's first equation. Total grams grow exponentially. The concentration x does not, because V grows too.

What β means: litres of feed per gram of new biomass. One gram needs 1/Y g of sugar, and each litre of feed delivers (feed sugar − s_0) usable grams.

Step 2 · Put s = s_0 into the substrate balance

With s constant, sV changes only because V does:

$$\frac{d(sV)}{dt} = s_0 \frac{dV}{dt} = s_0 \, v$$

Substitute, and take xV from Step 1:

$$v\,(s_f - s_0) = \frac{\mu_0}{Y_{x/s}} x_0 V_0 \, e^{\mu_0 t}$$

Solve for v. This is the slide's second equation:

$$v(t) = \beta \, \mu_0 \, x_0 V_0 \, e^{\mu_0 t} \qquad \beta = \frac{1}{Y_{x/s}\,(s_f - s_0)}$$

Explains original slide 15 · Section 9.2.2, Example 9.7

Step 3: integrate the feed to get V(t) and x(t)

Volume

Since dV/dt = v, integrate v(t) from Step 2:

$$V - V_0 = \int_0^t v\, dt' = \beta\, x_0 V_0\, (e^{\mu_0 t} - 1)$$

Divide by V_0 to get the slide's third equation:

$$\frac{V}{V_0} = 1 - \beta x_0 + \beta x_0\, e^{\mu_0 t}$$

Concentration: why it levels off

Divide total biomass by volume:

$$x = \frac{xV}{V} = \frac{x_0\, e^{\mu_0 t}}{1 - \beta x_0 + \beta x_0\, e^{\mu_0 t}}$$

At long times the exponentials dominate:

$$x \;\rightarrow\; \frac{1}{\beta} = Y_{x/s}\, (s_f - s_0)$$

Shortcut check: biomass made = yield × sugar used

$$xV - x_0 V_0 = Y_{x/s}\, (s_f - s_0)\, (V - V_0)$$

PART A · SLIDE 15 · WHEN O_2 RUNS OUT

Step 4: when O_2 caps growth, switch to a smaller k

Why growth hits a ceiling

The tank can transfer only so much O_2, which caps growth per litre (μx). With $\mu = \mu_o$, μx rises with x until it reaches the cap at biomass x_o':

$$\mu x \le q_{x,\max} = \mu_0 x_0'$$

From then on, run at the cap. Growth per litre is fixed, so total growth is proportional to volume:

$$\frac{d(xV)}{dt} = \mu_0 x_0' V$$

Deriving k

Sugar is eaten as fast as it arrives ($s \approx 0$), so feed = consumption:

$$v s_f = \frac{1}{Y_{x/s}} \frac{d(xV)}{dt} = \frac{\mu_0 x_0'}{Y_{x/s}} V$$

Divide by the feed sugar and use $v = dV/dt$:

$$\frac{dV}{dt} = k V, \qquad k = \frac{\mu_0 x_0'}{Y_{x/s} s_f}$$

$$V = V_1 e^{k(t - t_1)}, \qquad v = k V$$

Still an exponential feed, but with $k < \mu_o$: the slide's orange equation.

Explains original slide 15 · Section 9.2.2, Example 9.7

Checking the table on slide 15

Quantity	How it follows	Value
Max O_2 uptake	$kLa \times 0.9\ c^* = 650\ h^{-1} \times 0.9 \times 0.244$ mM	142.7 mmol L^{-1} h^{-1}
Cap on growth, q_x,max	O_2 cap ÷ O_2 needed per gram of yeast (book value)	5.12 g L^{-1} h^{-1}
Biomass at the cap, x_o'	$q_{x,max} \div \mu_o = 5.12 \div 0.25$	20.5 g L^{-1}
Time of switch	Step 3 $x(t) = 20.5$, with $\beta x_o \approx 0.021$, gives $V \approx 1.72\ V_o$	14.3 h
New exponent k	$\mu_o x_o' \div (Y \times 100\ g\ L^{-1})$, with $Y \approx 0.47\ g\ g^{-1}$	0.108 h^{-1}
Total time to $4V_o$	$14.3\ h + \ln(4 \div 1.72) \div 0.108 = 14.3 + 7.8$	≈ 22 h
Final biomass	$[x_oV_o + Y \times 100 \times (4V_o - V_o)] \div 4V_o$	≈ 35.8 g L^{-1} (slide: 35.6)

Y isn't printed on slide 15. About 0.47 g g^{-1} is what the slide's k and x_o' imply, and s_o is taken as ≈ 0. Rounding in Y explains 35.8 vs 35.6.

Exponential feed until O_2 runs out

$$x\,V = x_0 V_0\, e^{\mu_0 t}$$

$$v(t) = \beta\,\mu_0\,x_0 V_0\, e^{\mu_0 t}$$

$$\frac{V}{V_0} = 1 - \beta x_0 + \beta x_0\, e^{\mu_0 t}$$

$$\beta = \frac{1}{Y_{x/s}\,(s_f - s_0)}$$

Biomass grows exponentially, then levels off. When O_2 transfer caps growth at a biomass level x_0', keep feeding exponentially with a smaller exponent:

$$k = \frac{\mu_0\,x_0'}{Y_{x/s}\,s_f}$$

Baker's yeast, Example 9.7

$\mu_0 = 0.25\ \mathrm{h^{-1}},\ \ x_0 = 1\ \mathrm{g\,L^{-1}},\ \ s_f = 100\ \mathrm{g\,L^{-1}}$

$k_L a = 650\ \mathrm{h^{-1}},\ \ \text{harvest at } V = 4V_0$

Milestone	Value
O_2 limit reached at	x = 20.5 g L^{-1}
Time of switch	14.3 h
New feed exponent k	0.108 h^{-1}
Total time to $4V_0$	22 h
Final biomass	35.6 g L^{-1}

WORKED EXAMPLE 2 · PROBLEM

Designing an exponential feed

Given

Glucose fed at constant μ_0 to avoid by-products; no maintenance; residual glucose negligible.

Parameter	Symbol	Value
Initial volume	V_0	1 m³
Initial biomass	x_0	5 g L⁻¹
Glucose in feed	s_f	200 g L⁻¹
Biomass yield	$Y_{x/s}$	0.5 g g⁻¹
Growth rate	μ_0	0.2 h⁻¹

Find

(a) the initial feed rate v(0)

(b) the time to reach V = 2 m³

(c) the final biomass and the final feed rate

(d) the mass-transfer coefficient needed at the end, given:

$$Y_{O_2/x} = 0.684 \text{ mol } O_2 \text{ (C-mol X)}^{-1}, \quad 1 \text{ C-mol X} = 24.6 \text{ g}$$
$$c^*_{O_2} = 0.244 \text{ mM}, \quad \text{dissolved } O_2 \geq 10\% \text{ of saturation}$$

WORKED EXAMPLE 2 · SOLUTION

Volume doubles in 15 hours

(a) $\beta = \dfrac{1}{0.5 \times 200} = 0.01\ \mathrm{L\,g^{-1}}$

$v(0) = 0.01 \times 0.2 \times 5 \times 1000 = \mathbf{10}\ \mathrm{L\,h^{-1}}$

(b) $2 = 0.95 + 0.05\,e^{0.2t} \Rightarrow e^{0.2t} = 21$

$t = \ln 21 / 0.2 = \mathbf{15.2}\ \mathrm{h}$

(c) $x = 5 \times 21/2 = \mathbf{52.5}\ \mathrm{g\,L^{-1}}, \quad v = \mathbf{210}\ \mathrm{L\,h^{-1}}$

(d) $q_x = \mu_0\, x = 0.2 \times 52.5 = 10.5\ \mathrm{g\,L^{-1}\,h^{-1}}$

$r_{O_2} = \dfrac{10.5}{24.6} \times 0.684 = 0.292\ \mathrm{mol\,L^{-1}\,h^{-1}}$

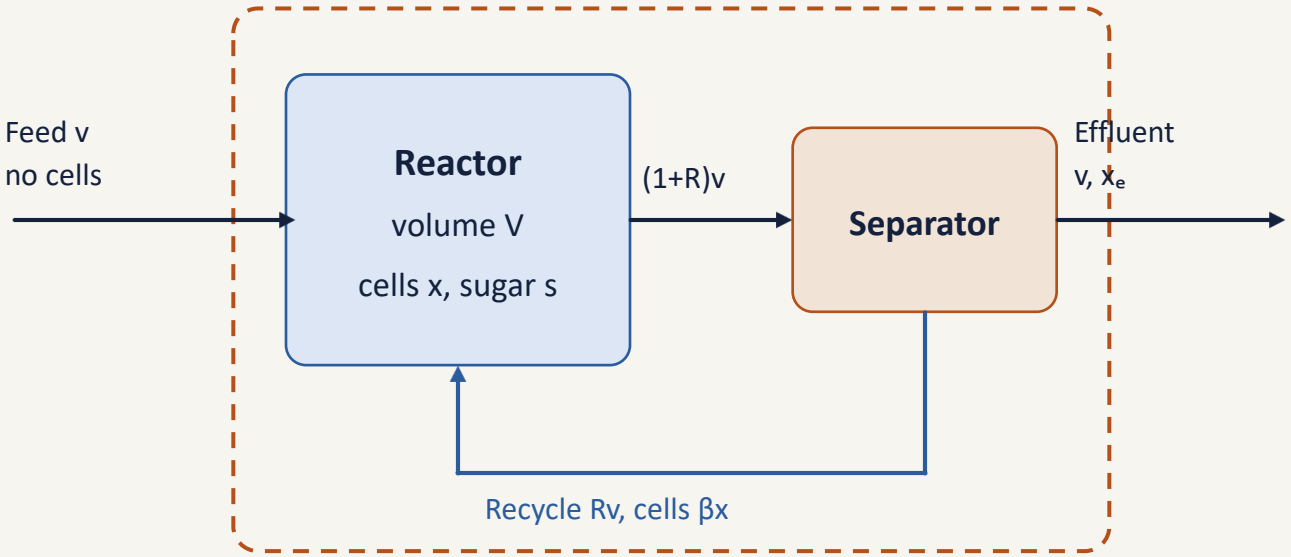
$k_L a = \dfrac{292\ \mathrm{mM\,h^{-1}}}{0.9 \times 0.244\ \mathrm{mM}} \approx \mathbf{1330}\ \mathrm{h^{-1}}$

t (h)	V (m³)	x (g L⁻¹)	v (L h⁻¹)
0	1.00	5.0	10
6	1.12	14.9	33
12	1.50	36.7	110
15.2	2.00	52.5	210

Too much oxygen demand

1,330 h⁻¹ exceeds a practical ~1,000 h⁻¹. At that limit O₂ caps growth at about 40 g L⁻¹ biomass (t ≈ 12.6 h): switch to the slower exponent k, or use O₂-enriched air or higher pressure.

Draw the balance around reactor + separator



The recycle crosses the boundary twice and cancels, so only the feed and the effluent count.

Definition

$$f = \frac{x_e}{x} < 1$$

Effluent cells relative to reactor cells. f < 1 because the separator holds cells back.

Assumptions

- Steady state, sterile feed
- Separator: no volume, no reaction
- Sugar s is the same in every stream
- D = v/V uses the fresh feed v

PART B · SLIDE 18 · THE IDEA

Two balances give μ = D·f and the outlet biomass

Biomass balance

In: none (sterile feed). Out: v·x_e. Made: μ·xV. At steady state, divide by xV:

$$0 = \mu\, xV - v\, x_e \qquad \mu = \frac{v}{V} \cdot \frac{x_e}{x} = Df$$

Cells only need to grow at a fraction f of the dilution rate, so D can exceed μmax without washout.

Monod then fixes s for any chosen f:

$$\mu_{\max} \frac{s}{K_s + s} = Df \;\Rightarrow\; s = \frac{K_s\, Df}{\mu_{\max} - Df}$$

Substrate balance

Sugar enters with the feed, leaves in the effluent at s, and is eaten at (μ/Y)·xV:

$$0 = v\,(s_f - s) - \frac{\mu}{Y_{x/s}}\, xV$$

Divide by V, use μ = Df, then cancel D:

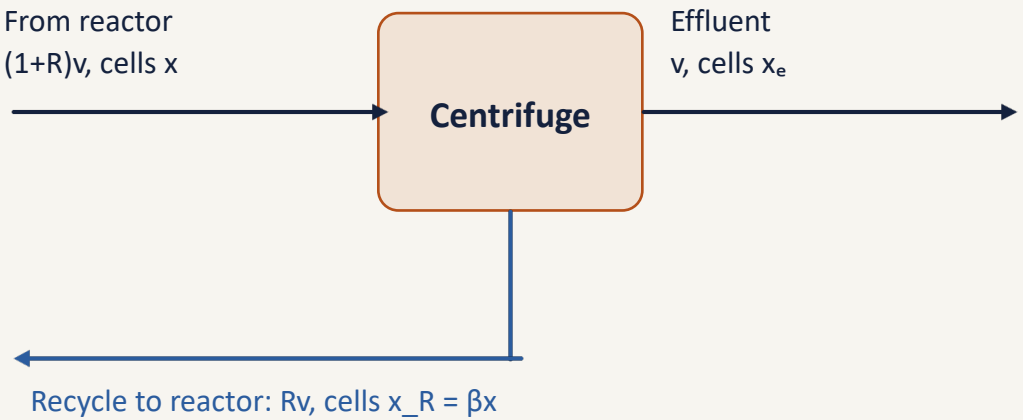
$$D\,(s_f - s) = \frac{Df\, x}{Y_{x/s}} \qquad x_e = f\, x = Y_{x/s}\,(s_f - s)$$

So the reactor holds 1/f times the effluent cells:

$$x = \frac{x_e}{f} = \frac{Y_{x/s}\,(s_f - s)}{f}$$

PART B · SLIDE 18 · OPTION A

Centrifuge: balance cells around the separator



Example 9.3 check

f = 0.276 and R = 2 give β = 1 + (1 − f)/R = 1.36: the recycle is 1.36 times thicker than the broth, and α = 2 × 1.36 ÷ 3 = 91 % of the cells go back.

Reactor outflow = feed + recycle = (1+R)v, at concentration x. Cells in = cells out:

$$(1+R)\,v\,x = R\,v\,(\beta x) + v\,x_e$$

Divide by v·x. This is the slide's result:

$$f = \frac{x_e}{x} = 1 + R - R\beta = 1 - R\,(\beta - 1)$$

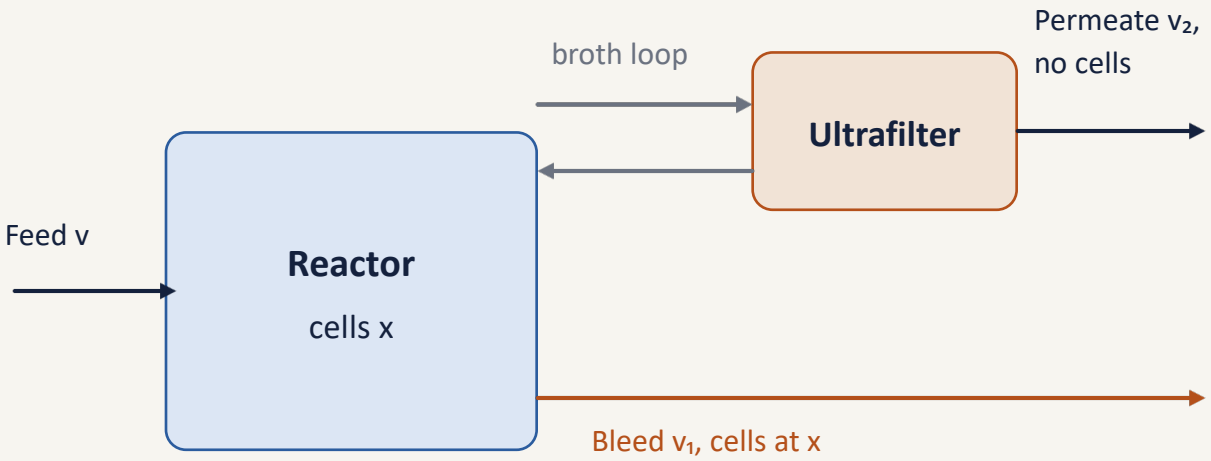
β > 1 measures how much the separator thickens the recycle. A larger R or β gives a smaller f.

Share of all cells sent back:

$$\alpha = \frac{R\,v\,\beta x}{(1+R)\,v\,x} = \frac{R\beta}{1+R}$$

Explains original slide 18 · Section 9.1.4, Example 9.3

Ultrafilter: cells leave only in the bleed



Biomass balance

Split the outflow into a bleed of broth (cells at x) and a cell-free permeate:

$$v = v_1 + v_2, \qquad \Omega = \frac{v_2}{v}$$

Only the bleed carries cells out, so at steady state:

$$0 = \mu\, xV - v_1\, x \;\Rightarrow\; \mu = \frac{v_1}{V} = D\,(1 - \Omega)$$

Same form as μ = Df, with f = 1 − Ω: the bleed fraction plays the role of f.

PART B · SLIDE 18 · OPTION B

Why productivity rises by 1/(1 – Ω)

Biomass in the reactor

Same whole-system sugar balance as before, now with μ = D(1 − Ω):

$$x = \frac{Y_{x/s}\,(s_f - s)}{1 - \Omega}$$

Cells build up 1/(1 − Ω) times higher than in a chemostat at the same sugar level s.

Productivity per litre

Biomass made per litre per hour:

$$P = \mu\, x = D\, Y_{x/s}\,(s_f - s)$$

A chemostat at the same μ and s makes the bracket below. Since D = μ/(1 − Ω):

$$P = \frac{1}{1 - \Omega}\,[\,\mu\, Y_{x/s}\,(s_f - s)\,]$$

Same yield, same conversion, but the tank treats 1/(1 − Ω) times more feed.

Same logic for the centrifuge: productivity rises by 1/f. All of the gain comes from the higher cell density in the reactor.

Keeping cells in lets D exceed μ

The idea

$f = x_e / x < 1$

$D\,f = \mu$

$x_e = Y_{x/s}\,(s_f - s)$

Substrate and product balances are unchanged. For any D, choosing f sets μ, and hence s.

A · Centrifuge or hydrocyclone

$f = 1 - R\,(\beta - 1)$

$R:$ recycle ratio, $\quad \beta = x_R / x$

β measures how much the separator thickens the recycle stream.

B · Ultrafilter with bleed

$\mu = D\,(1 - \Omega)$

$\Omega = v_2 / v:$ cell-free permeate fraction

Productivity rises by the factor 1/(1 – Ω).

Used in activated-sludge and anaerobic waste-water plants, membrane bioreactors, and perfusion cultures of animal cells.

Recycle rescues a too-small reactor

Given: $V = 0.5\ \mathrm{m^3}$, $v = 1\ \mathrm{m^3\,h^{-1}}$, $s_f = 4\ \mathrm{g\,L^{-1}}$, $\mu_{\max} = 1\ \mathrm{h^{-1}}$, $K_s = 1\ \mathrm{g\,L^{-1}}$, $Y_{x/s} = 0.5\ \mathrm{g\,g^{-1}}$

1 Without recycle: washout

$$D = \frac{v}{V} = 2\ \mathrm{h^{-1}} \;>\; D_{\max} = \frac{1 \times 4}{1+4} = 0.8\ \mathrm{h^{-1}}$$

2 Choose the effluent sugar level

$$s = 0.309\, s_f = 1.24\ \mathrm{g\,L^{-1}} \;\Rightarrow\; \mu = \frac{1.24}{1+1.24} = 0.553\ \mathrm{h^{-1}}$$

3 Share of cells allowed to leave

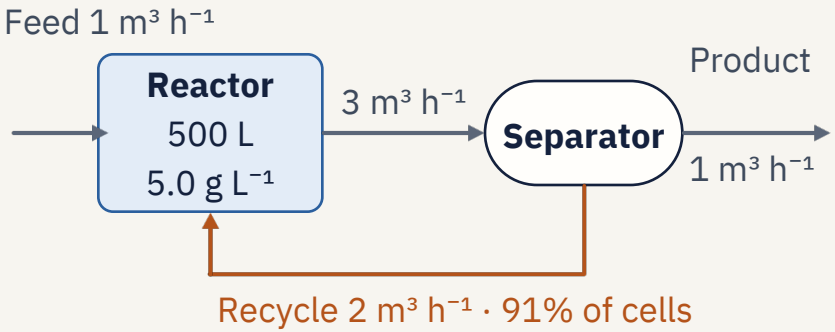
$$f = \frac{\mu}{D} = \frac{0.553}{2} = 0.276$$

4 Biomass out and inside

$$x_e = Y_{x/s}\,(s_f - s) = 1.38\ \mathrm{g\,L^{-1}}, \qquad x = \frac{x_e}{f} = 5.0\ \mathrm{g\,L^{-1}}$$

5 Productivity

$$q_x = D\,x_e = 2 \times 1.38 = \mathbf{2.76}\ \mathrm{g\,L^{-1}\,h^{-1}}$$



What the separator must do

$$\beta = 1 + \frac{1-f}{R} = 1 + \frac{0.724}{2} = 1.36$$

With recycle ratio R = 2, the recycle must be 1.36 times thicker than the broth.

Smaller reactor, bigger pump bill

Cheese whey, 15 m³ h⁻¹ with 4.5 wt% lactose, converted to lactic acid for polylactate. 10 m³ CSTR with an ultrafilter; 90% of the flow leaves as cell-free permeate (Ω = 0.9).

1 Cells leave only in the bleed

$$\mu = \frac{v_1}{V} = \frac{1.5}{10} = 0.15\ \mathrm{h^{-1}}, \qquad D = \frac{v}{V} = \frac{15}{10} = 1.5\ \mathrm{h^{-1}}$$

2 Tank needed without recycle

$$V_{\text{no recycle}} = \frac{v}{\mu} = \frac{15}{0.15} = 100\ \mathrm{m^3}$$

3 Membrane area at 10 L m⁻² h⁻¹

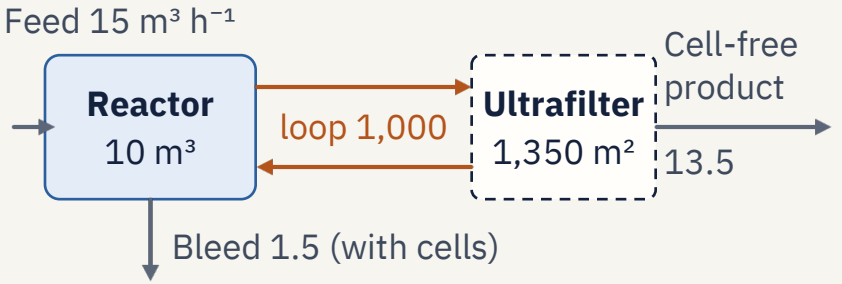
$$A = \frac{v_2}{J} = \frac{13.5\ \mathrm{m^3\,h^{-1}}}{0.010\ \mathrm{m^3\,m^{-2}\,h^{-1}}} = 1350\ \mathrm{m^2}$$

4 Cross-flow to keep it clean

$$Q \approx 0.74\ \mathrm{m^3\,m^{-2}\,h^{-1}} \times 1350\ \mathrm{m^2} \approx 1000\ \mathrm{m^3\,h^{-1}}$$

5 Pumping power, 4 bar, 65%

$$P = \frac{\Delta p\, Q}{\eta} = \frac{4 \times 10^5\ \mathrm{Pa} \times 0.278\ \mathrm{m^3\,s^{-1}}}{0.65} \approx \mathbf{171}\ \mathrm{kW}$$



Verdict

Recycle saves 90 m³ of tank but costs 171 kW of round-the-clock pumping and 1,350 m² of membrane. A cheap drum filter is needed anyway.

Balance gas transfer and liquid feed

Gas substrate S_1 (O_2, CH_4) limits

$$q_x = Y_{x/s_1}\, k_L a\, s_1^*$$

Liquid-fed substrate S_2 limits

$$q_x = D\, Y_{x/s_2}\, (s_{2,f} - s_2)$$

SCP from methane	D (h^{-1})	NH_3 feed (mg L^{-1})	$k_L a$ (h^{-1})	x (g L^{-1})	q_x (g L^{-1} h^{-1})
Base design	0.20	200	52	0.97	0.19
Target x = 10 g L^{-1}	0.20	1,449	535	10	2.0
Joint optimum	0.29	1,449	679	8.9	**2.54**

Design so both limits give the same productivity: extra mass-transfer capacity is wasted, and improving one feed forces you to improve the other.

WORKED EXAMPLE 3 · PROBLEM

Maintenance makes biomass peak

Given

A yeast grows on glucose in a CSTR; sterile feed, Monod kinetics, with maintenance.

Parameter	Symbol	Value
Max. growth rate	μ_{max}	0.6 h^{-1}
Saturation constant	K_s	0.25 g L^{-1}
Glucose in feed	s_f	20 g L^{-1}
True biomass yield	$Y_{x/s}^{true}$	0.5 g g^{-1}
Maintenance coefficient	m_s	0.05 g g^{-1} h^{-1}

Find

(a) s, x and observed yield at D = 0.05, 0.2, 0.5 h^{-1}

(b) the washout dilution rate

(c) roughly where biomass and productivity D·x peak

Hint: maintenance leaves s(D) unchanged — it only lowers the yield.

WORKED EXAMPLE 3 · SOLUTION

One third of the sugar is burnt at low D

(a) Observed yield sets x

$$s = \frac{K_s D}{\mu_{\max} - D}, \qquad \frac{1}{Y_{x/s}^{\text{obs}}} = \frac{1}{Y_{x/s}^{\text{true}}} + \frac{m_s}{D}$$

$$x = Y_{x/s}^{\text{obs}} \, (s_f - s)$$

D (h^{-1})	s (g L^{-1})	Y_{obs}	x (g L^{-1})	D·x (g L^{-1} h^{-1})
0.05	0.023	**0.333**	**6.66**	0.33
0.20	0.125	0.444	8.83	1.77
0.50	1.25	0.476	8.93	4.46

At D = 0.05 h^{-1}, Y_{obs} is only 2/3 of the true yield.

(b) Washout

$$D_{\max} = \frac{0.6 \times 20}{0.25 + 20} = \mathbf{0.593}\,\text{h}^{-1}$$

(c) Two different optima

$$x_{\max} \approx \mathbf{9.18}\,\text{g L}^{-1} \quad \text{at } D \approx 0.38\,\text{h}^{-1}$$

$$(Dx)_{\max} \approx \mathbf{4.59}\,\text{g L}^{-1}\text{h}^{-1} \quad \text{at } D \approx 0.535\,\text{h}^{-1}$$

Productivity peaks at 90% of washout (s = 2.04, x = 8.58 g L^{-1}).

WORKED EXAMPLE 4 · PROBLEM

Phenol in the feed: two steady states

Given

Phenol-degrading bacteria in a CSTR; phenol inhibits its own uptake:

$$\mu = \frac{\mu_{\max}\, s}{K_s + s + s^2/K_i}$$

Parameter	Symbol	Value
Max. growth rate	μ_{max}	0.5 h^{-1}
Saturation constant	K_s	0.5 g L^{-1}
Inhibition constant	K_i	8 g L^{-1}
Phenol in feed	s_f	30 g L^{-1}
Biomass yield	$Y_{x/s}$	0.45 g g^{-1}

Find

(a) the highest growth rate the culture can reach

(b) all steady states at D = 0.25 h^{-1}, and which are stable

(c) what happens if the tank is started full of feed ($s = s_f$)

(d) the D giving maximum productivity on the stable branch

Hint: stability needs $d\mu/ds > 0$ (slide 24).

WORKED EXAMPLE 4 · SOLUTION

Run left of the peak — and start up in batch

(a) Peak of μ

$$s = \sqrt{K_i K_s} = \sqrt{4} = 2.0\,\mathrm{g\,L^{-1}}$$

$$\mu = \frac{0.5 \times 2}{0.5 + 2 + 0.5} = \mathbf{0.333}\,\mathrm{h^{-1}}$$

(c) Start-up at s_f

$$\mu(30) = \frac{15}{143} = 0.105 < D$$

Washout is stable too.

(d) Maximum productivity

$$s_{\mathrm{opt}} = 1.67\,\mathrm{g\,L^{-1}}$$

$$D = 0.332\,\mathrm{h^{-1}}$$

$$x = 12.75\,\mathrm{g\,L^{-1}}$$

$$q_x = \mathbf{4.23}\,\mathrm{g\,L^{-1}h^{-1}}$$

That is 99.7% of μ_{peak} — one phenol pulse from washout.

At D = 0.25: 3.31 g L^{-1} h^{-1}, 22% less but safe.

(b) Steady states at D = 0.25 h^{-1}

$$s^2 - 8s + 4 = 0 \;\Rightarrow\; s = 4 \pm \sqrt{12}$$

Steady state	s (g L^{-1})	x (g L^{-1})	Stable?
Left branch	**0.536**	**13.26**	Yes
Right branch	7.46	10.14	No

SUMMARY

Six things to remember

1

The CSTR is the most productive reactor; its optimum D sits close to washout and trades yield for rate.

2

Inhibition and maintenance shift the optimum and lengthen the residence time.

3

Coupled tanks and cell recycle add robustness and capacity, at a cost in pumping and control.

4

Match gas–liquid mass transfer to the liquid-phase rate; neither should be wasted.

5

Fed-batch avoids overflow and repression: feed exponentially at constant μ until O_2 limits.

6

Check stability and competitors; finish low-s effluents in a PFR after the CSTR.